

PIVOT TABLE

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In [33]: # A pivot table is used to summarize and analyse data by grouping rows and
#applying aggregate functions such as sum,mean,count,max,min,etc....

#step1: Load dataset
import pandas as pd
df = pd.read_csv("people-100.csv")
print(df.head())
print(df.columns)

#step2:Create Age Column
#convert str to datea
df['Date of birth'] = pd.to_datetime(df['Date of birth'])
today = pd.Timestamp.today()
df['Age'] = (
    (today - df['Date of birth']).dt.days // 365
)
print(df[['First Name','Sex','Age']].head())

#1.Basic Pivot Table
#Average age by Gender
pivot1 = pd.pivot_table(
    df,
    values='Age',
    index='Sex'
)
print(pivot1)

#2.pivot table with aggfunc='mean',
pivot2 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='sum'
)
print(pivot2)

#3.pivot table with aggfunc='count',
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pivot3 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='count'
)
print(pivot3)

#4.multiple aggregations
pivot4 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc=['mean', 'max', 'min']
)
print(pivot4)

#5.Adds Overall Average
pivot5 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='mean',
    margins=True
)
print(pivot5)

#6.Group by job title and gender
#Average age by jb title and Gender.
pivot6 = pd.pivot_table(
    df,
    values='Age',
    index='Job Title',
    columns='Sex',
    aggfunc='mean'
)
print(pivot6)

#count people by job and gender
pivot7 = pd.pivot_table(
    df,

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values='User Id',  
index='Job Title',  
columns='Sex',  
aggfunc='count',  
fill_value=0  
)  
print(pivot7)
```

	Index	User Id	First Name	Last Name	Sex	\
0	1	88F7B33d2bcf9f5	Shelby	Terrell	Male	
1	2	f90cD3E76f1A9b9	Phillip	Summers	Female	
2	3	DbeAb8CcdfefC2c	Kristine	Travis	Male	
3	4	A31Bee3c201ef58	Yesenia	Martinez	Male	
4	5	1bA7A3dc874da3c	Lori	Todd	Male	

		Email	Phone	Date of birth	\
0		elijah57@example.net	001-084-906-7849x73518	1945-10-26	
1		bethany14@example.com	214.112.6044x4913	1910-03-24	
2		bthompson@example.com	277.609.7938	1992-07-02	
3		kaitlinkaiser@example.com	584.094.6111	2017-08-03	
4		buchananmanuel@example.net	689-207-3558x7233	1938-12-01	

	Job Title
0	Games developer
1	Phytotherapist
2	Homeopath
3	Market researcher
4	Veterinary surgeon

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Index(['Index', 'User Id', 'First Name', 'Last Name', 'Sex', 'Email', 'Phone',
      'Date of birth', 'Job Title'],
      dtype='object')

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	First Name	Sex	Age
0	Shelby	Male	80
1	Phillip	Female	116
2	Kristine	Male	33
3	Yesenia	Male	8
4	Lori	Male	87

	Age
Sex	
Female	64.981132
Male	54.957447

	Age
Sex	
Female	3444
Male	2583

	Age
Sex	
Female	53
Male	47

	mean	max	min
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      Age  Age Age
Sex
Female 64.981132 117 8
Male   54.957447 117 5
      Age

```

```

Sex
Female 64.981132
Male   54.957447
All     60.270000

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```

Sex                Female  Male
Job Title
Accountant, chartered management  95.0  88.0
Accounting technician             17.0   NaN
Advertising account executive     18.0   NaN
Agricultural consultant           NaN  34.0
Air broker                        84.0   NaN
...                               ...   ...
Travel agency manager             NaN 117.0
Veterinary surgeon                NaN  87.0
Warden/ranger                     NaN  45.0
Warehouse manager                 NaN 101.0
Waste management officer          NaN  10.0

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[91 rows x 2 columns]
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Sex                Female  Male
Job Title
Accountant, chartered management    1    1
Accounting technician               1    0
Advertising account executive       1    0
Agricultural consultant             0    1
Air broker                          1    0
...                               ...   ...
Travel agency manager               0    1
Veterinary surgeon                  0    1
Warden/ranger                       0    1
Warehouse manager                   0    1
Waste management officer             0    1

```

```
[91 rows x 2 columns]
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In []: